38 Carbon-14 has a proton number (Z) of 6 and a nucleon number (A) of 14.

Nitrogen-14 has a proton number of 7 and a nucleon number of 14.

Carbon-14 emits β-particles to form nitrogen-14.

Which nuclide equation describes this process?

A $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{1}\beta$

B $^{6}_{14}C \rightarrow {}^{7}_{14}N + {}^{1}_{0}\beta$

C $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}\beta$

D $^{6}_{14}C \rightarrow {}^{7}_{14}N + {}^{-1}_{0}\beta$